ALGEBRAIZATION OF A FORMAL FINITE AND PROPER ALGEBRAIC SPACE OVER A COMPLETE NOETHERIAN BASE

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Editorial note. Counted span is the author's own proof environment, source lines 10216-10236, immediately following the statement at 10193-10214.

Editorial note. The proof uses Grothendieck's existence theorem twice: once in the reformulated form to produce the module, once as full faithfulness to algebraize it; both are navigation pointers into the same chapter.

Editorial note. The final step setting Y equal to the relative spectrum is displayed and completed inside the span.

Allowed standard prerequisites (editorial).

- Thickening property for cartesian squares of morphisms. chapter lemma lemma-thicken-property-morphisms-cartesian, cited at proof line 10220. Shows each vertical map f_n : Y_n -> X_n is finite because its base change to X_1 equals the given finite f_1.
- Reformulation of Grothendieck's existence theorem. chapter remark remark-reformulate-existence-theorem, cited at proof line 10226. Produces the coherent O_X-module F restricting to the given F_n on each X_n and records that its support is proper over S.
- Grothendieck's existence theorem: the completion functor is fully faithful. chapter theorem theorem-grothendieck-existence, cited at proof line 10231. Lets the compatible multiplication maps on the F_n descend to an algebra structure on F.
- Relative spectrum of a sheaf of algebras. used at proof line 10235. Realises the algebraized object as Y = Spec_X(F), which is finite over X and proper over S.

Source reference. Original lemma 09ZZ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2044–2044. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 08BF: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10085–10085. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 08BE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9990–9990. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let A be a Noetherian ring complete with respect to an ideal I. Write $S = \mathrm{Spec}(A)$ and $S_n = \mathrm{Spec}(A/I^n)$. Let $X \to S$ be a morphism of algebraic spaces that is separated and of finite type. For $n \geq 1$ we set $X_n = X \times_S S_n$. Suppose given a commutative diagram*

$$\begin{array}{ccccccc} Y_1 & \longrightarrow & Y_2 & \longrightarrow & Y_3 & \longrightarrow & \ldots \\ \downarrow & & \downarrow & & \downarrow & & \\ X_1 & \xrightarrow{i_1} & X_2 & \xrightarrow{i_2} & X_3 & \longrightarrow & \ldots \end{array}$$

of algebraic spaces with cartesian squares. Assume that

(1) *$Y_1 \to X_1$ is a finite morphism, and*
(2) *$Y_1 \to S_1$ is proper.*

Then there exists a finite morphism of algebraic spaces $Y \to X$ such that $Y_n = Y \times_S S_n$ for all $n \geq 1$. Moreover, Y is proper over S.

Proof. Let's write $f_n : Y_n \to X_n$ for the vertical morphisms. As the squares in the statement are cartesian we see that the base change of f_n to X_1 is f_1. Thus Lemma 09ZZ shows that f_n is a finite morphism. Set $\mathcal{F}_n = f_{n,*}\mathcal{O}_{Y_n}$. Using that the squares are cartesian we see that the pullback of $\mathcal{F}_{n+1}$ to X_n is $\mathcal{F}_n$. Hence Grothendieck's existence theorem, as reformulated in Remark 08BF, tells us there exists a coherent $\mathcal{O}_X$-module $\mathcal{F}$ whose restriction to X_n recovers $\mathcal{F}_n$. Moreover, the support of $\mathcal{F}$ is proper over S. As the completion functor is fully faithful (Theorem 08BE) we see that the multiplication maps $\mathcal{F}_n \otimes_{\mathcal{O}_{X_n}} \mathcal{F}_n \to \mathcal{F}_n$ fit together to give an algebra structure on $\mathcal{F}$. Setting $Y = \underline{\mathrm{Spec}}_X(\mathcal{F})$ finishes the proof. □